

Introduction, objectives and overview of the overall research programme

Despite rapid advances, the current state of AI in healthcare remains correlation-driven, fragmented and poorly adapted to clinical reality. Most deployed models operate as statistical predictors rather than decision technologies and therefore struggle with external validity, transportability and regulatory acceptance. Systematic reviews confirm that adoption is slowed by fragmented data environments, limited explainability and a lack of prospective clinical evidence. In cardiovascular applications, the literature reveals broad enthusiasm but few implementation case studies and no clear pathways for compliance with emerging AI regulations. This leaves hospitals, regulators and patients with "AI as a promise" rather than a demonstrably safe and auditable intervention tool.

The proposal is timely because emerging regulatory and policy frameworks now require trustworthy, explainable and interoperable AI in high-risk healthcare settings. PROJECT-X directly responds to these policy drivers while contributing to broader digital-health and societal objectives through fair, robust AI tools that are clinically actionable across diverse populations.

PROJECT-X's overarching ambition is:

1. to replace correlation-centric models with causal objectives and counterfactual evaluation;
2. to treat interoperability and governance as first-class design constraints;
3. to embed human-in-the-loop co-creation and targeted stakeholder training;
4. to execute healthcare-embedded case studies that generate evidence on safety, adoption and value.

Together, these elements define a cohesive research programme addressing key skills and knowledge gaps in healthcare AI.

Specific measurable research and innovation objectives

O1 – Trustworthy causal AI (WP1)

Develop explainable, fair and sensitivity-aware causal AI models that articulate assumptions, quantify uncertainty and provide counterfactual “what-if” and cause-effect reasoning at both patient and policy levels.

Success KPIs

- KPI1.1: ≥ 1 multimodal AI model integrated with causal modules.

- KPI1.2: ≥ 10 clinicians test the tool and $\geq 80\%$ report improved confidence in AI-supported decisions.
 - KPI1.3: ≥ 1 joint report or white paper citing causal evidence generated within the project.
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O2 – Interoperability for identifiability (WP2)

Establish multimodal, robust and privacy-preserving data flows and the semantic structures required by target causal graphs.

Implement federated identifiability and transportability checks across sites to ensure that models remain valid, generalisable and clinically deployable under heterogeneous real-world conditions.

Success KPIs

- KPI2.1: successful integration of ≥ 4 data modalities into a unified semantic schema.
 - KPI2.2: ≥ 1 pilot model per use case integrating ≥ 3 formalised clinical or physiological rules.
 - KPI2.3: ≥ 1 data exchange protocol between academic and industrial partners.
 - KPI2.4: ≥ 1 published guideline or white paper reviewed by technical, clinical and regulatory stakeholders.
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O3 – Awareness and co-creation (WP3)

Deliver role-specific training for clinicians, data stewards and regulators; implement human-in-the-loop AI tools; and co-create interfaces with patients to improve usability, trust and real-world alignment.

Success KPIs

- KPI3.1: clinician and patient usability ratings $\geq 80\%$.
 - KPI3.2: ≥ 2 co-design workshops with ≥ 15 participants.
 - KPI3.3: ≥ 3 AI tools supporting interactive clinician feedback and retraining mechanisms.
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O4 – Clinical adaptation and evidence (WP4)

Translate AI-based decision support into healthcare practice by identifying practical barriers to deployment.

Conduct retrospective evaluations across:

- Rare Neurodegenerative Cohorts
- Structural Heart Intervention Cohorts
- Heart Failure Cohorts
- population-level studies
- routine clinical workflows

Success KPIs

- KPI4.1: ≥ 1 health-economic or cost-effectiveness evaluation.
 - KPI4.2: ≥ 2 policy briefs or white papers.
 - KPI4.3: implementation of at least one AI system in an additional clinical environment.
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Cross-cutting objective (CO)

Operationalise relevant AI regulations and best-practice guidelines across all objectives by embedding:

- transparency
- risk classification
- robustness
- human oversight
- lifecycle monitoring

throughout model development and deployment.

Success KPIs

- Unified compliance and risk-management framework.
- ≥ 1 regulatory-readiness toolkit publication.
- Dissemination through at least two international channels.

DC1 – Physiology-aware AI for clinical decision support

Host: Research-intensive European university with expertise in biomedical signal processing and explainable AI (WP1, WP2, WP4)

Objectives

- Formalise clinical and physiological knowledge including heuristics, rules, causal relations and constraints for cardiovascular use cases.
- Develop compact physiology-aware AI architectures that integrate domain knowledge at data, model and decision levels.
- Provide transparent explanations and uncertainty estimates that clinicians can interrogate.

Expected Results

- Knowledge representations and software prototypes for physiology-aware AI across multiple imaging and biosignal modalities.
- Demonstration of improved robustness and generalisability compared with black-box baselines.
- Reusable modules supporting explainability and trustworthy AI across the network.

Key Secondments

- University medical centre specialising in cardiovascular clinical workflows and healthcare data infrastructures.
- Engineering school with expertise in causal AI, medical imaging and multimodal analysis.
- Clinical imaging centre providing advanced cardiovascular imaging datasets.

Contribution to the Network

Provides the physiological and mechanistic foundations supporting trustworthiness, interoperability and clinical adaptation.

DC2 – Causal multimodal integration for cardiovascular risk prediction

Host: Technical university with expertise in medical imaging, multimodal AI and causal modelling (WP1, WP2, WP4)

Objectives

- Identify causal estimands across time series, laboratory measurements and biomarkers.

- Develop architectures supporting temporal causal reasoning and counterfactual inference.
- Design multimodal fusion approaches that improve interpretability and robustness.

Expected Results

- Causality-informed risk prediction algorithms.
- Temporal causal models for clinical decision support.
- Benchmarking studies demonstrating improved robustness and interpretability.

Key Secondments

- Academic AI centre specialising in fairness analysis and explainability.
- Clinical cardiology environment supporting real-world validation.

Contribution to the Network

Provides core multimodal causal modelling methods used throughout the network.

DC3 – AI-ready clinical data and processing environments

Host: University medical centre with expertise in medical informatics and health-data infrastructures (WP2, WP3, WP4)

Objectives

- Assess organisational and technical AI readiness of healthcare environments.
- Co-design readiness frameworks with clinicians, IT professionals, patients and governance experts.
- Implement and evaluate a regulatory-compliant AI deployment pipeline.

Expected Results

- Data-readiness assessment framework.
- Infrastructure-readiness assessment framework.
- Guidelines for practical deployment of trustworthy healthcare AI.
- Open-source demonstrator environment.

Key Secondments

- Digital-health research centre specialising in health-data spaces and implementation science.

- Industrial AI environment focusing on explainability and uncertainty.
- Clinical implementation site enabling cross-institutional evaluation.

Contribution to the Network

Bridges AI development, governance and healthcare deployment.

DC4 – Sensitivity and uncertainty modelling for trustworthy AI

Host: Research and technology organisation with expertise in explainable AI and computational physiology (WP1, WP3, WP4)

Objectives

- Develop methods for quantifying prediction uncertainty and sensitivity.
- Create clinician-oriented confidence metrics.
- Evaluate the impact of uncertainty information on clinical decision-making.

Expected Results

- Integrated uncertainty and sensitivity framework.
- Calibration and robustness metrics.
- Evidence of improved interpretability and trust.

Key Secondments

- Academic centre specialising in explainable AI and evaluation methodologies.
- Medical-device company developing AI-enabled decision-support technologies.
- Clinical validation environment.

Contribution to the Network

Strengthens trustworthiness by making uncertainty explicit and clinically actionable.

DC5 – Privacy-preserving embedded AI for contactless physiological monitoring

Host: Technical university specialising in biomedical measurement technologies (WP2, WP4)

Objectives

- Analyse privacy and regulatory implications of contactless physiological monitoring.

- Develop distributed and privacy-preserving AI architectures.
- Evaluate privacy-performance trade-offs.
- Create privacy-by-design guidance for regulated medical technologies.

Expected Results

- Distributed privacy-preserving sensing platform.
- Benchmarking framework for privacy, efficiency and performance.
- Regulatory guidance for trustworthy embedded AI.

Key Secondments

- Clinical environment supporting physiological monitoring applications.
- Medical-device manufacturer developing advanced monitoring systems.

Contribution to the Network

Provides privacy-preserving deployment methodologies relevant to sensitive healthcare settings.

DC6 – Human-in-the-loop causal AI for physiological signals and medical imaging

Host: Technical university with expertise in explainable AI and biosignal analysis (WP1, WP3, WP4)

Objectives

- Design and evaluate human-in-the-loop AI paradigms.
- Strengthen transparency and user control.
- Improve safety and trust in clinical decision support.

Expected Results

- General human-in-the-loop framework.
- Multiple validated application demonstrators.
- Quantitative evidence regarding usability and trust.

Key Secondments

- Clinical medical-informatics environment.
- Emergency and intensive-care device manufacturer.
- Company specialising in long-term physiological monitoring technologies.

Contribution to the Network

Operationalises human oversight within trustworthy AI systems.

DC7 – Trustworthiness monitoring for digital health services

Host: University with expertise in digital health, health informatics and human-computer interaction (WP3, WP4)

Objectives

- Define continuous trustworthiness-monitoring methodologies.
- Design dashboards and monitoring tools.
- Assess clinical, organisational and economic impact at scale.

Expected Results

- Trustworthiness-monitoring framework.
- Monitoring dashboards and validation studies.
- Recommendations for large-scale deployment.

Key Secondments

- AI research laboratory supporting explainability and monitoring integration.

Contribution to the Network

Makes trustworthiness measurable throughout the AI lifecycle.

DC8 – Causal machine learning and counterfactual analysis for interventional medicine

Host: University research institute focusing on AI in cardiovascular medicine (WP1, WP2, WP4)

Objectives

- Investigate treatment-outcome confounding in interventional healthcare.
- Perform multicentre federated model development and evaluation.
- Develop counterfactual and sensitivity analysis methods.

Expected Results

- Causal multimodal prediction models.

- Multicentre evaluation framework.
- Open-source tools for explanation and counterfactual analysis.

Key Secondments

- University medical centre specialising in clinical data engineering and federated infrastructures.
- Clinical partner supporting implementation in routine practice.

Contribution to the Network

Demonstrates trustworthy causal AI in a high-impact interventional healthcare setting.

DC9 – Clinical adaptation and prognostic modelling for rare neurological disease

Host: Academic medical centre with expertise in digital health and rare-disease research (WP1, WP3, WP4)

Objectives

- Develop multimodal causal prognostic models.
- Identify causal connections among physiological, clinical and biomarker trajectories.
- Evaluate usability and clinical impact.

Expected Results

- Interpretable multimodal prognostic model.
- Benchmark dataset and validation framework.
- Clinical and regulatory evaluation report.

Key Secondments

- AI-driven cardiovascular medicine research institute.
- University medical centre supporting federated evaluation and workflow integration.

Contribution to the Network

Demonstrates transferability of trustworthy AI methodologies beyond cardiovascular medicine into complex neurological disease management.

Pertinence and innovative aspects of the research programme

Global health and policy roadmaps call for AI systems that are safe, interoperable and implementable in real-world healthcare environments. Yet adoption consistently stalls because of privacy concerns, interoperability challenges, bias, trust deficits and implementation barriers. PROJECT-X addresses these challenges through **causal objectives, governance-ready engineering and healthcare-embedded validation**, directly targeting gaps identified by healthcare-system stakeholders and implementation-science studies.

The programme responds to calls for organisation-wide strategies and enabling infrastructures capable of supporting trustworthy AI deployment beyond isolated research pilots. It also aligns with recent analyses emphasising resilience to variability, uncertainty and distribution shifts in healthcare data.

By embedding regulatory requirements such as AI governance frameworks, medical-device regulations, data-protection requirements and interoperability standards into the research process, the project goes beyond conventional AI prototypes and instead develops **conformity-ready documentation, monitoring pipelines and implementation frameworks** suitable for reuse by healthcare providers and industrial stakeholders.

Beyond the state of the art: scientific and technical innovations

The proposed innovations build upon and extend recent advances in causal explainability, federated learning, multimodal AI and trustworthy clinical decision support.

1. Causal, multimodal and federated AI

The programme develops multimodal causal models and federated transportability methodologies capable of supporting robust effect estimation across institutions.

The research integrates:

- longitudinal healthcare records,
- medical imaging,
- physiological measurements,
- biomarker information,
- patient-generated health data,

into semantically harmonised and privacy-preserving analytical frameworks.

Unlike classical predictive AI, the project explicitly considers transportability, identifiability and causal validity across heterogeneous clinical environments.

2. Resilient explainability and uncertainty quantification

The project develops novel methodologies for:

- sensitivity analysis,
- uncertainty quantification,
- counterfactual explainability,
- human-in-the-loop interactions.

These capabilities are integrated directly into explainable AI pipelines rather than added as post-processing components. Their purpose is to address reproducibility, robustness and trust limitations frequently reported in healthcare AI deployments.

3. Knowledge-augmented and physiology-aware models

The programme develops physiology-aware and knowledge-integrated AI architectures that incorporate formalised scientific and clinical knowledge into learning systems.

These approaches aim to:

- improve interpretability,
- improve out-of-distribution robustness,
- improve data efficiency,
- reduce model complexity,
- support deployment on constrained computing environments.

Such models directly address limitations of purely data-driven deep learning in healthcare applications.

Beyond the state of the art: clinical adaptation, governance and awareness

The programme deliberately moves beyond the traditional goal of producing incrementally better predictive models.

Instead, it develops a **complete healthcare AI pathway**, covering:

- algorithms,
- interoperability infrastructures,
- governance mechanisms,
- clinical adoption,
- regulatory readiness,
- implementation evaluation.

Key innovations include:

Healthcare-embedded validation

The programme performs healthcare-embedded case studies that progress from retrospective analyses toward shadow-mode and implementation-oriented evaluations.

Evaluation criteria include:

- diagnostic accuracy,
- workflow impact,
- time-to-decision,
- adoption indicators,
- implementation feasibility.

Stakeholder co-creation

The project systematically involves:

- clinicians,
- patients,
- data stewards,
- healthcare organisations,
- regulatory stakeholders,

through co-design activities intended to address adoption barriers such as:

- organisational resistance,
- workflow disruption,
- trust concerns,
- perceived unfairness,

- transparency requirements.

Regulatory science integration

The programme integrates legal, regulatory and implementation expertise to produce:

- regulatory-aligned documentation,
- monitoring frameworks,
- impact-assessment methodologies,
- implementation guidance.

These outputs are intended to support future standardisation and policy-development activities.

Positioning versus existing doctoral programmes

Many existing doctoral programmes in AI and digital health focus predominantly on:

- algorithms,
- data infrastructures,
- ethics,
- management.

Few programmes integrate all of these dimensions into a single coherent framework.

PROJECT-X distinguishes itself through three characteristics:

Four-pillar architecture

The programme is organised around:

1. Trustworthiness
2. Interoperability
3. Awareness
4. Clinical adaptation

This common framework links all doctoral projects and training activities.

Regulatory integration by design

Regulatory and governance requirements are embedded directly into research questions, infrastructures and evaluation strategies instead of being treated as downstream compliance activities.

Combination of causal AI and real-world implementation

The programme combines:

- advanced causal machine learning,
- healthcare implementation science,
- interoperability research,
- human-centred design,
- health-economic evaluation,

within a single doctoral training environment.

This combination addresses a recognised shortage of specialists capable of simultaneously understanding:

- causal modelling,
- healthcare workflows,
- data governance,
- validation methodologies,
- regulatory compliance.

Expected transformation

By the end of the programme, the consortium will deliver:

1. Reference methodologies for causal estimation, explainability, uncertainty analysis and sensitivity assessment.
2. Interoperable and privacy-preserving data and computation frameworks.
3. A new generation of researchers trained at the intersection of AI, healthcare and governance.
4. Demonstrated clinical adaptation workflows supported by implementation and compliance frameworks.

Collectively, these outputs establish a blueprint for **regulation-aligned causal AI in healthcare**, moving beyond predictive modelling toward safe, trustworthy and effective decision-support systems suitable for real-world deployment.

1.2. Soundness of the proposed methodology

Overall methodology

We adopt a highly interdisciplinary and strongly collaborative approach that prioritises early and open exchange of ideas within the consortium and with the wider community.

Our overarching ambition is to replace correlation-driven pipelines with causal AI systems that are resilient to real-world data challenges, interoperable across heterogeneous healthcare infrastructures and regulatory compliant.

The programme balances fundamental research with applied validation in healthcare environments, ensuring that methods remain theoretically rigorous, technically auditable and practically implementable in routine healthcare.

This framework is underpinned by four methodological advances.

Causal AI and counterfactual reasoning

Instead of relying solely on black-box prediction, the programme develops models capable of estimating cause-effect relationships and counterfactual outcomes.

The models explicitly encode assumptions and quantify both structural and statistical uncertainty.

This directly addresses important reproducibility, robustness and bias challenges identified in healthcare AI.

Regulatory-grade pipelines

Compliance with healthcare AI regulations and governance frameworks is treated as a methodological requirement rather than a post hoc constraint.

Risk classification, transparency, robustness, human oversight and lifecycle monitoring are embedded throughout the development process within a quality-managed framework.

The methodology operationalises major regulatory requirements relevant to high-risk healthcare AI systems, including medical-device, data-governance and AI-governance obligations.

Federated and privacy-preserving interoperability

The programme leverages federated learning, split learning, standardised healthcare data models and selection-diagram methodologies to transform identifiability and privacy into engineering problems.

This directly addresses long-standing interoperability and data-sharing barriers present in healthcare IT ecosystems.

The approach seeks to ensure that privacy preservation, semantic interoperability and model validity are jointly addressed rather than treated as independent challenges.

Healthcare-embedded implementation and co-creation

Healthcare case studies are designed in close collaboration with:

- clinicians,
- healthcare IT professionals,
- patients,
- healthcare organisations,
- regulatory stakeholders.

This enables the programme to address organisational, ethical and workflow barriers that frequently hinder AI adoption.

Together, these methodological pillars address the principal barriers limiting healthcare AI impact:

- data quality,
- interoperability,
- transparency,
- trust,
- governance,
- regulation,
- implementation.

The resulting closed-loop methodology follows the sequence:

Causal Question



Identifiability Analysis



Federated Data Preparation



Model Development



Clinical Evaluation



Assurance & Documentation

and is consistently applied across all research projects and healthcare use cases.



Integration of methods and disciplines to pursue the objectives

Achieving the objectives requires integration of technical, clinical, regulatory and social-science disciplines.

Computer science and causal inference

Development of novel algorithms for:

- robust training,
- sensitivity analysis,
- counterfactual evaluation,
- fairness-aware learning,

grounded in modern causal-inference and trustworthy-AI research.

Biomedical and clinical sciences

Validation of models within cardiovascular and related healthcare domains ensures that:

- causal graphs,
- estimands,
- predictions,
- recommendations

reflect clinically meaningful mechanisms and healthcare-relevant endpoints.

This addresses the scarcity of rigorously evaluated healthcare implementation studies.

Regulatory science and legal expertise

The methodology aligns with:

- healthcare AI regulations,
- medical-device requirements,
- interoperability obligations,

- data-protection regulations.

Research activities include co-development of:

- conformity-ready documentation,
 - risk-management artefacts,
 - monitoring protocols,
 - compliance assessment methodologies.
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Social sciences and human-factors research

The methodology incorporates:

- participatory design,
- organisational-change models,
- human-centred assessment,
- ethical evaluation.

Trust, usability and perceived fairness are explicitly evaluated as measurable outcomes rather than secondary effects.

Interdisciplinarity embedded across work packages

Interdisciplinarity is built directly into the work-package structure.

Trustworthiness work package

Combines:

- machine learning,
- statistics,
- ethics,
- explainability,
- causal reasoning.

Interoperability work package

Combines:

- health informatics,

- systems engineering,
- privacy technologies,
- legal compliance.

Awareness work package

Combines:

- human-centred design,
- training sciences,
- stakeholder engagement,
- adoption science.

Clinical adaptation work package

Combines:

- biomedical sciences,
- implementation science,
- organisational change,
- regulatory evaluation.

The same interdisciplinary integration is reflected within individual doctoral projects.

Examples include:

- causal machine learning integrated with clinical decision support,
- AI-readiness assessment combining informatics and regulatory science,
- human-in-the-loop AI combined with social-science approaches to trust and adoption,
- disease-specific clinical applications combined with federated learning and causal inference.

These projects collectively exemplify interdisciplinary collaboration in healthcare AI.

Gender dimension and diversity aspects

Sex, gender and additional diversity characteristics are treated as scientific variables.

These variables are incorporated throughout model development and evaluation:

- represented in causal structures as potential causes, effects or mediators,
- included within datasets whenever available,
- analysed via subgroup performance assessments,
- monitored through calibration and error analysis,
- incorporated into bias-detection and mitigation workflows.

Human-centred design activities involve diverse groups of clinicians and patients to ensure that explanations and decisions remain understandable and trustworthy across populations.

Integration of diversity considerations strengthens:

- validity,
- robustness,
- fairness,
- generalisability.

Open science practices

Open science is implemented within the constraints imposed by medical-data protection and high-risk AI governance.

All publications will be open access wherever possible.

A data-management strategy ensures adherence to FAIR principles through:

- standard ontologies,
- persistent identifiers,
- machine-readable metadata,
- reproducible workflows,
- documentation standards.

The programme intends to release, where legally and ethically feasible:

- generic algorithms,

- benchmarking frameworks,
- software components,
- synthetic datasets.

Key findings will be disseminated through:

- publications,
- preprints,
- analysis plans,
- reusable software assets.

The guiding principle throughout the project is:

As open as possible, as closed as necessary.

1.3. Quality and credibility of the training programme

1.3.1 Overview and content structure of the doctoral training programme

The training programme equips doctoral candidates with the integrated expertise needed to design, evaluate and deploy causal AI systems in real clinical and regulatory environments, while systematically integrating transferable skills, inter- and multidisciplinary content, and intersectoral exposure in line with the MSCA “triple-i” principles.

Current doctoral training in AI and digital health primarily focuses on predictive modelling and technical implementation. Only a limited number of programmes address causal inference, clinical evaluation and compliance with evolving European frameworks, such as the AI Act, the Medical Device Regulation (MDR), and the European Health Data Space (EHDS), in an integrated way.

The Doctoral Network (DN) provides a structured training framework that combines:

1. Individual research and local doctoral training at the host institution, including enrolment in an accredited PhD programme and discipline-specific coursework;
2. Network-wide scientific and transferable skills training, delivered through dedicated schools, workshops and online modules; and

3. Intersectoral and international secondments in hospitals, industry and regulatory/standardisation bodies, where candidates apply causal AI methods in real-world health settings.

This architecture is implemented and monitored via the Training and Career Development work package and the Training Programme Coordination task in the management work package.

Together, local doctoral-school provision and the network-wide programme form a single, coherent curriculum: each DC's Personal Career Development Plan (PCDP) maps how local and network-wide activities, including secondments, contribute to TG1–TG6 and to the four programme pillars (trustworthiness, interoperability, awareness, clinical adaptation).

1.3.2 Training objectives and stakeholder needs

The training programme is guided by the following network-wide objectives, which were developed in dialogue with the network's key stakeholder groups (including university hospitals, technology and data providers, SMEs, policy and standardization bodies) and are reflected in the individual DC project descriptions and work package training goals:

TG1 – Core scientific skills in causal and trustworthy AI for health

DCs will develop an in-depth understanding of causal modelling, counterfactual reasoning, transportability and identifiability, sensitivity and uncertainty analysis, fairness and robustness, multimodal data fusion and interoperable implementation in clinical contexts.

TG2 – Clinical, biomedical and public health literacy

DCs will develop comprehensive expertise in cardiovascular and related clinical domains, as well as in study design, outcomes, care pathways and health system constraints. This expertise will enable them to collaborate with healthcare professionals in defining clinically meaningful estimands and interpreting model outputs.

TG3 – Data governance, regulation and ethics

Training covers GDPR and health data governance, EU AI Act requirements for high-risk medical AI systems, MDR/IVDR interfaces, the European Health Data Space, research integrity and responsible research and innovation (RRI), as well as patient and right-holder perspectives.

TG4 – Transferable and complementary skills

In line with MSCA guidance and the European Competence Framework for Researchers, DCs will develop competencies in project management, teamwork, leadership, intercultural collaboration, scientific writing and communication, stakeholder engagement, open science, and teaching and mentoring.

TG5 – Innovation, entrepreneurship and impact

DCs will learn how to translate research into innovation, including IPR management, standardization, needs and market analysis, and co-creation with industry, hospitals and public bodies.

TG6 – Career development and resilience

DCs will be prepared for diverse careers in academia, hospitals, industry, regulatory agencies and public administration through targeted career-development training, structured exposure to non-academic environments, and mentoring.

These six training objectives (TG1–TG6) are operationalized through the integrated training architecture and triple-i implementation described in Section 1.3.3.

Each training objective (TG1–TG6) directly supports specific components of the research programme: TGs 1–3 align with methodological WPs 1–3, TG2 reinforces clinical adaptations, TG4 supports management activities, and TG5–6 underpin the impact, innovation and career-development goals embedded in the training and impact work packages.

1.3.3 Triple-I approach

The programme implements a systematic, cross-cutting “triple-i” approach to achieve TG1–TG6:

International

All projects involve cross-border recruitment and at least one international secondment. Network-wide events rotate between countries and are streamed or recorded to maximize accessibility. DCs are encouraged and supported to present at international conferences and participate in international research networks.

Intersectoral

Secondments and co-supervision connect universities with university hospitals, digital health SMEs, and other non-academic partners. DCs gain first-hand experience of

clinical workflows, regulatory compliance processes and product-development lifecycles, which fosters a shared language between themselves, clinicians and regulators.

Interdisciplinary

Each DC is exposed to computer science, statistics, medicine, law/regulation and human-centred design/social sciences through joint courses, co-supervision and cross-work-package collaborations, reflecting the sociotechnical nature of trustworthy causal AI in healthcare.

1.3.4 Special training elements of the network

Local training at host organizations

Each host institution provides structured doctoral training through its graduate or doctoral school. This training includes core scientific courses and seminars, as well as modules on transferable skills such as research ethics, gender and diversity, and open science. Wherever possible, these courses are made available on-site or online to DCs from other beneficiary institutions, thereby increasing efficiency and coherence. Individual research projects, group meetings, journal clubs, and local clinical seminars ensure that DCs are fully integrated into vibrant research and clinical environments.

Network-wide training

Non-academic partners lead or co-lead sessions on clinical workflows, regulatory compliance, entrepreneurship and innovation management, ensuring that intersectoral exposure is substantive and aligned with stakeholder needs. These activities, with indicative timing and contributions of beneficiaries and associated partners, are listed in Table 1.

Table 1. Main Network-Wide Training Events, Conferences and Contribution of Beneficiaries

1. Kick-Start training week organised by the coordinating institution

Day 1 (TG1–2): public lectures from invited European and international experts and consortium partners on topics such as translational health research, biomedical device development, wearable monitoring, cardiovascular disorders, signal processing, model-based AI, medical imaging, mobile health and explainable AI.

Day 2 (TG4–5): project, scientific and training-goal overview, theoretical and practical introductions, industry session.

Day 3 (TG4–5–6): transferable skills course and career planning; scientific communication and presentation skills; open-science basics with a focus on research-data management (RDM), FAIR principles, ethics and research integrity.

Day 4 (TG5): poster session and laboratory visits.

2. Summer School for “Succeeding in my PhD Journey”

- Pre-assignments: personal study-plan draft, career scenarios (TG6)
- Visiting professors, former MSCA alumni career stories, groupwork (TG4)
- Post-assignments: PCDP completion, groupwork (TG6)

3. Training on ethics and regulations for AI in healthcare

Day 1 (TG3): tutorials on EU regulations and ethics, and their implications for healthcare AI models.

Day 2 (TG1–3): privacy challenges, data protection, split-learning and federated-AI methodologies.

Day 3 (TG5): how to translate scientific results into innovation, including the pathway from research to product development.

4. Training on interoperability standards for clinical AI models

Day 1 (TG3): public lectures from invited international experts and consortium partners on interoperability standards.

Day 2 (TG3): semantic and syntactic metadata standards for AI models.

Day 3 (TG2–5): integrating large-scale data and explainable AI into daily clinical practice; hospital and data-centre visits; lectures on regulatory standards.

5. International minisymposium on Trustworthy AI for Healthcare

The consortium will organise a dedicated minisymposium on trustworthy AI for healthcare at major international conferences. The event will feature scientific presentations by doctoral fellows and supervisors, showcasing project results to a

global audience. The minisymposium will foster networking with international experts, enhance collaboration opportunities, and increase the project’s impact, visibility and scientific reach.

6. Course on AI and Cardiovascular Health

Day 1 (TG4–6): clinical decision-support tools empowered by AI.

Day 2 (TG1–4–6): role and impact of trustworthy, causal and explainable AI principles in cardiology.

Day 3 (TG4–5): unlocking insights from cardiology data using AI and exploring opportunities for human-AI collaboration.

7. Final International Conference on Trustworthy AI for Healthcare

Day 1 (TG1–2): public lecture on translating AI research into clinical practice; invited lectures.

Day 2–3 (TG1–2–5): scientific talks by fellows and contributed talks by external participants.

Day 4 (TG4–6): focus session on future collaborations and sustainability beyond project duration; networking and institutional visits.

1.4. Quality of the supervision (including mandatory joint supervision for industrial and joint doctorate projects)

1.4.1 Qualifications and supervision experience of supervisors

Supervisor Profile	Publications & Academic Standing	Supervision Experience	Expertise & Research Environment
Supervisor A Associate Professor, Technical University, PhD	H-index ~13, ~80 publications	3+ years of supervision; 1 completed PhD; 1 ongoing PhD; primary supervisor of a doctoral candidate	Medical data science, explainable AI, biosignal processing, clinical AI validation. Access to high-performance computing facilities and healthcare technology infrastructure.

Supervisor Profile	Publications & Academic Standing	Supervision Experience	Expertise & Research Environment
Supervisor B Associate Professor, Coordinating University, PhD & Habilitation	H-index ~16, ~80 publications	5+ years of supervision; completed and ongoing PhD supervision; doctoral-network coordinator and work-package leader	Biomedical signal processing, ECG/EEG analysis, model-based AI, explainable AI, causal machine learning. Access to AI laboratories, GPU infrastructure and strong healthcare collaborations.
Supervisor C Consultant Clinician & Associate Professor, Academic Medical Centre	H-index ~35, ~155 publications	8+ years of supervision; multiple completed and ongoing PhDs	Clinical electrophysiology, digital health, AI implementation, cardiovascular medicine. Access to longitudinal registries, multimodal physiological datasets and advanced clinical facilities.
Supervisor D Chair Professor in Biomedical Engineering	H-index >60, >800 publications (combined laboratory output)	More than 50 completed doctoral supervisions within associated laboratory environment	Biomedical measurement technology, contactless monitoring, AI-enabled medical systems, physiological sensing technologies. Access to advanced monitoring laboratories and specialised device-development environments.
Supervisor E Senior Scientist in Biomedical Engineering	Active publication record with multiple ongoing	4+ years supervisory experience; multiple ongoing PhD candidates	Camera-based monitoring, physiological signal processing, embedded AI, privacy-preserving healthcare technologies. Access to translational engineering

Supervisor Profile	Publications & Academic Standing	Supervision Experience	Expertise & Research Environment
	doctoral projects		laboratories and clinical validation environments.
Supervisor F Professor of Biomedical Engineering, Research & Technology Organisation	H-index ~33, >130 publications	10+ years supervision; approximately 10+ completed PhDs; multiple ongoing doctoral candidates	Causal AI, explainability, computational physiology, cardiovascular modelling. Access to dedicated AI-development teams, curated datasets and translational healthcare research facilities.
Supervisor G Professor of Medical Informatics	H-index ~25, >220 publications	10+ years supervision; multiple completed and ongoing PhDs	Medical informatics, interoperability, health-data infrastructures, AI validation, biosignal analytics. Access to secure health-data infrastructures, federated-learning environments and clinical information systems.
Supervisor H Professor of Biomedical Engineering and Medical Imaging	H-index ~57, >500 publications	Approximately 30 years supervision; >35 completed PhDs	Medical imaging, mobile health, trustworthy AI, explainable AI and multimodal learning. Access to large-scale computing facilities and advanced imaging laboratories.
Supervisor I Professor in Digital Health	H-index ~25-30; >100 publications	~25 years supervision;	Digital health, healthcare transformation, human-centred technology,

Supervisor Profile	Publications & Academic Standing	Supervision Experience	Expertise & Research Environment
		multiple ongoing doctoral candidates	sustainable healthcare systems. Access to digital-health innovation environments and advanced communication testbeds.
Supervisor J Senior Researcher, MD-PhD-MBA	H-index ~15; ~80 publications	5+ years supervision; multiple ongoing doctoral candidates	Medical informatics, digital therapeutics, healthcare implementation science and clinical AI adoption. Access to health-data spaces and healthcare innovation ecosystems.
Supervisor K Full Professor & Director of AI Institute in Healthcare	H-index ~23, ~140 publications	Multiple completed and ongoing doctoral candidates	Federated learning, multimodal AI, cardiovascular medicine, imaging and biosignal analytics. Access to large multicentre datasets, clinical partners and hospital IT infrastructures.

1.4.2 Quality of supervision for DN

Supervision in the DN is organised around a multi-layered structure that combines day-to-day guidance at the host institution with network-wide monitoring and support:

- Each DC has a main academic supervisor at the recruiting beneficiary and at least one co-supervisor or mentor from another sector (hospital, SME or other non-academic partner). Where appropriate, an additional mentor from a complementary discipline (e.g. law/regulation, human-centred design) is included. Together, they form an Individual Training Panel (ITP) or supervisory committee.

- Within the first three months, the ITP and DC jointly agree a Personal Career Development Plan (PCDP), specifying research objectives, training needs, secondments, open-science commitments and career goals. The PCDP is reviewed at least every six months and updated when needed; key milestones are aligned with host doctoral-school requirements and network deliverables, and PCDP updates are validated through the Supervisory Board as part of training monitoring activities.

Day-to-day supervision follows a clear and consistent policy across the consortium:

- Regular meetings between DCs and their main supervisors (typically every 2–4 weeks) focus on scientific progress, skills development and well-being.
- ITP meetings (in person or online) are held at least twice per year, often aligned with network-wide events, to review PCDPs, training progress, secondment planning and career perspectives.
- Supervisors operate an open-door communication culture, encouraging DCs to raise scientific or personal concerns early. Short ad hoc meetings and online interactions complement formal supervision sessions and provide a two-way feedback channel.

Progress and review procedures are embedded in project governance:

- The Supervisory Board (SB) oversees the overall supervision strategy and ensures that PCDPs, secondments and training objectives are implemented consistently across sites, in line with the MSCA Guidelines on Supervision.
- Training and supervision quality are monitored through annual progress reports, PCDP updates and anonymous feedback questionnaires completed by DCs after major training events.
- Clear procedures for conflict prevention and resolution are included in the Consortium Agreement (e.g. two-step escalation from local resolution within the ITP to intervention by the SB and, if needed, an external ombudsperson), ensuring that problems can be addressed in a timely and confidential manner.

Supervision explicitly includes career development, integration and well-being. DCs are supported in building professional networks (e.g. conference participation, joint publications, involvement in standards or guideline activities) and are encouraged to co-supervise MSc students or contribute to teaching, under guidance. Integration into local research and clinical groups is fostered through lab meetings, journal clubs and clinical

seminars, ensuring that each DC works within a vibrant environment rather than in isolation.

Although the project is a standard Doctoral Network (not an Industrial Doctorate or Joint Doctorate scheme), each DC benefits from this structured co-supervision team, which provides complementary intersectoral and interdisciplinary guidance comparable in quality to mandatory joint-supervision schemes in industrial and joint doctorates.

2.1. Contribution to structuring doctoral training at the European level and to strengthening European innovation capacity, including the potential for: Meaningful contribution of the non-academic sector to the doctoral training, as appropriate to the implementation mode and research field

2.1.1 Structuring doctoral training at the European level

The Doctoral Network will establish a pan-European training ecosystem for causal AI in healthcare, integrating academic, clinical, industrial and regulatory stakeholders into a coherent framework. It contributes directly to the MSCA objective of structuring doctoral education by embedding a common curriculum, interoperable learning modules and shared infrastructures across universities, hospitals, technology and data providers, and policy or regulatory bodies.

In contrast to existing AI or digital-health programmes, the network uniquely combines causal inference, clinical translation and regulatory alignment, such as the EU AI Act and the European Health Data Space, positioning Europe as a leader rather than a late adopter in trustworthy medical AI. This bridges computer science, legal-ethical governance and clinical practice, creating graduates who can operate AI systems under EU regulatory, ethical and clinical standards.

Each Doctoral Candidate (DC) follows a unified curriculum combining AI, medical translation, data-governance and regulatory training (see Section 1.3).

The network promotes international, intersectoral and interdisciplinary mobility. DCs enrol in accredited programmes in EU Member States or Associated Countries, receiving co-supervision from academic and non-academic sectors.

All DCs will complete a substantial secondment (4-6 months) in another sector, typically hosted by non-academic partners (hospitals, SMEs, large industry or regulatory/standardisation bodies). Supervisors from these organisations sit on the Supervisory Board and Individual Training Panels (ITP). This ensures exposure to the non-academic sector, enhances employability, provides complementary supervision and creates networking opportunities.

Cross-sectoral and cross-national mobility is further supported by international secondments between clinical and data-science environments, creating long-term institutional links between academic and non-academic partners and building a pipeline of intersectoral research careers.

Measurable outcomes include at least three joint doctoral courses, three transferable-skills modules and the majority ($\geq 80\%$) of secondment months in non-academic settings.

2.1.2 Strengthening European innovation capacity

The project will strengthen Europe's capacity to safely and ethically innovate with AI in healthcare, directly addressing key policy priorities such as the EU AI Act and the European Health Data Space. The consortium's innovation pathway focuses on four mutually reinforcing elements:

Translational pipelines for regulatory-grade AI.

By operationalising risk-based classification, transparency, robustness, human oversight and post-market monitoring within doctoral projects, the network establishes a training-by-doing model for regulatory compliance.

Interoperable and FAIR infrastructures.

Doctoral projects use FHIR-based semantic interoperability standards and privacy-preserving infrastructures (e.g. medical data integration centres) to enable secure cross-border data exchange and reproducible experiments.

Public-private knowledge transfer.

Hospitals, SMEs and technology providers co-define research problems, host secondments and co-supervise DCs, ensuring that methods are grounded in real clinical, product-development and regulatory needs.

Socio-technical innovation awareness.

Training modules are co-designed with experts in ethics, law, human-computer interaction and social sciences so that DCs learn to build human-in-the-loop systems that are not only technically advanced but societally trustworthy and aligned with patient and citizen expectations.

These elements directly reinforce European innovation capacity and prepare DCs for non-academic careers across med-tech, hospitals, regulatory bodies and digital-health start-ups.

Developing sustainable (= lasting) elements of doctoral programmes after the end of the DN funding

2.1.3 Sustainability and long-term impact

To ensure impact beyond the project, the consortium will embed core modules in a **European Causal AI Academy** (sanitised: **European AI Training Academy**), a virtual hub for shared curriculum, case studies and regulatory toolkits. Partners integrate successful modules into their doctoral institutions (e.g. ECTS credits, network schools in regular course catalogues), sustaining supervision, secondment and training.

The AI knowledge hub, aligned with standards such as ResearchComp and ESG, serves as a Horizon Europe reference point and disseminates materials on AI Act and EHDS implementation. An Europe-wide alumni hub and an Innovation and Sustainability Committee will track indicators such as joint publications, IP, follow-up projects and external users of academy resources, and will adapt the programme where needed.

After the end of DN funding, the consortium will jointly target further funding for research and training (e.g. MSCA Staff Exchanges, EIT initiatives, Erasmus+, Horizon Europe, COST actions and comparable European programmes).

The network, training programme and European AI Training Academy provide a strong basis for cooperation, ensuring that the project's effect on training and innovation in Europe outlasts the project itself.

2.2. Credibility of the measures to enhance the career perspectives and employability of researchers and contribution to their skills development

Career development is a key priority, anchored in the Training and Career Development work package. Within their first three months, each Doctoral Candidate (DC) establishes a Personal Career Development Plan (PCDP) with their supervisory team, detailing research milestones, scientific and transferable-skills targets, secondments, teaching and outreach contributions, and short-, medium- and long-term career objectives. PCDPs are reviewed annually and used to adapt training and mobility activities to evolving interests and labour-market trends.

Every fellow is supported by a PhD committee comprising at least one academic supervisor, one non-academic or clinical supervisor and, where relevant, a representative from the secondment host. This multi-mentor structure provides consistent guidance and impartial advice on academic, clinical, industrial and policy careers, and facilitates early identification of additional skills or networking needs. Training progress and PCDPs are monitored through dedicated reporting and structured feedback mechanisms.

Combining methodological innovation in causal and trustworthy AI with real-world constraints such as data quality, workflow integration, interoperability and regulatory documentation prepares graduates to lead multidisciplinary teams in hospitals, industry R&D departments, start-ups and research infrastructures. This unique profile is difficult to obtain in single-discipline programmes and significantly enhances employability.

Triple-I exposure and professional networks

Triple-I exposure (international, intersectoral and interdisciplinary) increases DCs' visibility, experience with clinical, product-development and regulatory workflows, and their ability to combine technical, clinical, legal and human factors in practice.

Transferable skills, innovation, entrepreneurship and regulatory literacy

The transferable-skills programme embedded in the training and impact work packages equips DCs with strong competences in:

- project and time management;
- teamwork and leadership;
- scientific and non-technical communication;
- intercultural collaboration;
- stress and conflict management.

Training in IP protection, entrepreneurship, innovation management and the use of a Research-to-Innovation (R2I) canvas enables fellows to identify and articulate the scientific, technological and societal value of their work.

Positioning for diverse and sustainable career paths

These elements position DCs for sustainable careers:

- in academia and research infrastructures as experts in causal and trustworthy AI for health;
- in hospitals and healthcare providers as clinical data scientists and digital-health leaders;
- in industry and start-ups as specialists in trustworthy, regulation-ready AI;
- as advisors to regulatory agencies, policy bodies and professional societies.

The network's long-term AI knowledge hub will provide a lasting platform for showcasing fellows' outputs and profiles and maintaining an alumni network, thereby extending the impact of the training beyond the project lifetime and supporting continued cross-sector mobility.

2.3. Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

Plan for the dissemination and exploitation activities, including communication activities

Strategic framework and governance

Dissemination, exploitation and communication (DEC) activities are implemented under a coordinated governance structure, ensuring coherence with research, innovation and sustainability actions and systematic monitoring of impact indicators.

Scientific dissemination and open science

Scientific results will be disseminated primarily in the fields of computer science, artificial intelligence and healthcare. Prestigious Q1-D1 journals will be targeted across both technical and medical domains. Top-tier AI conferences will also be pursued.

Each doctoral candidate is expected to produce at least three peer-reviewed publications or conference contributions during the project, yielding a minimum of approximately 27 scientific outputs.

Dissemination follows open-science principles to maximise visibility, transparency and reproducibility. Publications will be made openly accessible via institutional repositories and the European Open Science Cloud (EOSC), with data and code released under permissive licences whenever possible, following the principle of "as open as possible, as closed as necessary."

Technical documentation and reproducibility reports will accompany significant software or algorithmic releases, supporting trustworthy AI assets aligned with European regulatory requirements.

Joint sessions at major international conferences, dedicated symposia on trustworthy AI in healthcare and cross-sector workshops involving clinical and regulatory stakeholders will ensure that project results reach both academic and applied communities.

Knowledge transfer and exploitation of results

Exploitation measures will transform scientific achievements into innovations, guidelines and policy inputs. Each doctoral project will include a **Research-to-Innovation Canvas (R2I Canvas)** that maps:

- potential results,
- intellectual property,
- target users,
- technology readiness,
- exploitation pathways.

Potential exploitation routes include:

- clinical deployment,
- integration into healthcare infrastructures,
- contributions to standards and guidelines,
- commercial development,
- open tools and reference implementations.

Result maturity and exploitation pathways will be monitored through the project governance structure to ensure alignment with industrial needs, clinical relevance and policy priorities.

Long-term sustainability and post-project exploitation

A virtual European AI training and knowledge hub will ensure continuity of training activities, open access to educational materials and uptake of successful modules beyond project completion. Educational content may be integrated into partner curricula and professional training programmes.

The hub and its associated events will provide a long-term interface with policy and regulatory stakeholders, ensuring that project outputs continue to contribute to implementation of emerging European AI and health-data regulations as well as future doctoral training initiatives in digital health.

Strategy for the management of intellectual property and foreseen protection measures

Through the integration of open science, structured knowledge transfer, sustainable exploitation and active communication, the Doctoral Network will create a European reference point for training in trustworthy AI, support innovation and policy development in health AI, and strengthen public trust in responsible AI in healthcare.

The IP management strategy underpins these goals.

IP management will follow Horizon Europe requirements and will be implemented through:

- a Consortium Agreement,
- an IP Management Plan,
- dedicated exploitation and innovation structures with Technology Transfer Office (TTO) participation.

The framework governs:

- ownership,
- access rights,
- confidentiality,
- exploitation routes.

Background intellectual property remains with the provider. Project results belong to the generating partner(s) or are jointly owned under appropriate agreements. Access rights to background and results are granted royalty-free for project implementation and research activities, while commercial use follows fair and reasonable conditions.

Beneficiaries will protect commercially relevant results through appropriate intellectual-property instruments. Other outputs, such as:

- reference implementations,
- benchmarks,
- selected software tools,

may be released under open licences to maximise uptake and reuse.

Regular IP reviews will identify protectable outcomes and recommend appropriate exploitation routes, including:

- industrial valorisation,

- licensing,
- spin-outs and start-ups,
- open dissemination,
- integration into long-term training and knowledge infrastructures.

Doctoral candidates will receive dedicated training in:

- patenting,
- licensing,
- spin-out creation,
- technology transfer,
- balancing IP protection with open-science requirements.

Together, these measures provide a credible, proportionate and well-governed framework for dissemination, exploitation, communication and intellectual-property management that is fully aligned with MSCA expectations.

2.4. The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Expected scientific impact(s)

The project will strengthen the scientific basis of trustworthy and interpretable AI in healthcare by moving beyond purely correlation-driven prediction towards evidence-based and clinically deployable approaches aligned with European regulatory frameworks.

Three major scientific advances are expected:

Advanced modelling and validation methodologies

The network will develop methods that combine transparent modelling, identifiability assessment and cross-site validation mechanisms to establish robust and transferable results across institutions and datasets. Expected outcomes include improved model robustness, generalisability and evidence quality compared with baseline approaches.

Explainability, uncertainty and human-centred AI

New methods will be developed to quantify uncertainty, evaluate model robustness

under changing conditions and support human oversight through interactive decision-support tools. These approaches are expected to improve clinician confidence, transparency and decision-support quality in real-world healthcare settings.

Regulatory integration and interoperability

The project will align AI-development lifecycles with relevant European regulatory and interoperability frameworks by developing documentation templates, evaluation procedures, risk-management approaches and interoperability standards suitable for healthcare applications. This work will result in unified compliance approaches, regulatory-readiness toolkits and evaluations across multiple clinical environments.

The programme will generate open-access scientific publications, FAIR datasets and reusable software resources that support wider adoption of trustworthy healthcare AI and contribute to European health-data and interoperability initiatives.

Expected economic and technological impact(s)

The project will strengthen knowledge transfer between data science, clinical practice and the digital-health sector by producing validated and interoperable AI methods that reduce development risks and support adoption by hospitals, SMEs and technology providers.

Impacts are expected through three complementary pathways:

Innovation in clinical decision support

Clinical use cases in cardiology, telemonitoring and related domains will demonstrate more robust, interpretable and efficient AI-supported solutions and provide foundations for future adoption in additional healthcare areas.

Interoperable digital-health infrastructures

Alignment with recognised interoperability and data-sharing standards will support development of reusable digital-health infrastructures and architectures that can be adopted across institutions and countries.

Human capital and innovation ecosystems

The project will train a cohort of specialists with expertise at the intersection of AI, healthcare, governance and implementation, strengthening Europe's competitiveness and innovation capacity in digital health.

By reducing barriers between research, validation and implementation, the project is expected to accelerate the path from scientific discovery to deployment-ready

healthcare solutions and improve returns on investments in healthcare innovation.

Expected societal impact(s)

The project will contribute to a safer, more transparent and more accountable healthcare ecosystem supported by responsible digital technologies.

Benefits for patients and citizens

Integration of AI-supported tools into healthcare workflows is expected to improve diagnostic accuracy, support personalised prevention and decision-making, reduce delays in care and strengthen transparency while maintaining privacy and accountability. The project also aims to support critical understanding of AI-supported recommendations among both healthcare professionals and end users.

Healthcare-system resilience and sustainability

Cross-country case studies will generate practical implementation blueprints addressing governance, risk management, workforce engagement and organisational adoption. These outputs are expected to improve efficiency, reduce unnecessary procedures and help healthcare systems use resources more effectively.

Policy and governance uptake

The project will provide evidence, guidelines and practical tools that support policymakers, regulators, standards organisations and healthcare leaders in deploying trustworthy digital health solutions. This contributes to public confidence in data-driven healthcare and supports Europe's leadership in responsible and ethical AI innovation.

Taken together, these scientific, economic and societal impacts support European digital-health and AI strategies while contributing to broader objectives related to health, innovation, infrastructure and sustainable healthcare transformation.

3. Quality and Efficiency of the Implementation

3.1. Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

WP Number: 1

WP title: Trustworthy, Explainable and Causal AI

Start month – End month: M1 – M36

Lead participant

Academic Beneficiary A

Participants

Academic Beneficiaries B, C, D, E

DCs involved

1,2,4,6,8

Objectives:

This WP will ensure that AI models are transparent, interpretable and able to quantify uncertainty, thereby enhancing trustworthiness and supporting compliance with applicable governance requirements in operational settings.

Task 1.1 - Fairness analysis of state-of-the-art AI models in specific populations:

Underrepresented groups are often insufficiently represented in training data, leading to dataset shifts and performance disparities. We will perform a detailed fairness analysis of advanced AI models for decision-support applications. This includes identifying proxy variables encoding sensitive attributes through explainability methods and validating results using composite metrics balancing fairness and performance. This task will establish baseline benchmarks for subsequent developments.

Task 1.2 – Model Explainability and Interpretability:

Develop population-level explainability methods for multimodal data by creating unbiased and generalisable interpretability frameworks for safety-critical decision support. We will quantify modality contributions and assess feature relevance through attribution methods combined with counterfactual explanations.

Task 1.3 – Sensitivity and Uncertainty Analysis:

Analyse sensitivity and quantify uncertainty in multimodal AI systems. Integrate uncertainty estimates into user-facing outputs to support safe human oversight and to identify unfamiliar populations, environments or operating conditions.

Task 1.4 – Causality in Multimodal AI:

Investigate causal relationships and confounding variables that may influence both interventions and outcomes. Develop causal modelling approaches capable of

improving robustness, transparency and interpretability in complex multimodal environments.

WP Number: 2

WP title: Interoperable, Robust and Privacy-Preserving AI Systems

Start month – End month: M1 – M36

Lead participant

Academic Beneficiary F

Participants

Academic Beneficiaries C, G, A

DCs involved

1,2,3,5,9

Objectives:

The aim is to develop interoperable, privacy-preserving, robust and knowledge-aware AI systems that comply with regulatory requirements and integrate into operational information infrastructures.

Task 2.1 – Integration of Multimodal Information:

Develop machine-learning methods to integrate heterogeneous data sources for improved prediction, classification and risk assessment. Multimodal fusion architectures, temporal models, harmonisation strategies and augmentation pipelines will be investigated. Causal reasoning methods will be combined with advanced representation-learning approaches to generate interpretable outputs.

Task 2.2 – Robust and Knowledge-Augmented Learning:

Improve robustness and generalisability across heterogeneous datasets by integrating formalised domain knowledge into learning architectures. Beyond model-agnostic approaches, we will develop knowledge-aware models that embed mathematical, logical and causal relations while reducing complexity and improving interpretability.

Task 2.3 – Privacy-Preserving AI Models:

Develop and evaluate privacy-preserving machine-learning methods for embedded and distributed environments. Implement distributed learning approaches combining local processing and centralised inference while minimising data exposure. Evaluate privacy, resource efficiency and performance trade-offs, and develop privacy-by-design recommendations aligned with applicable regulations.

Task 2.4 – Data, Processing and Model Interoperability:

Identify, evaluate and select interoperability standards for data, models and processing environments. This includes semantic and syntactic metadata standards, secure communication protocols, privacy assessments and deployment requirements. Interoperability gaps will be identified together with developers, users and governance stakeholders.

WP Number: 3

WP title: Human-AI Collaboration

Start month – End month: M1 – M36

Lead participant

Academic Beneficiary B

Participants

Academic Beneficiaries H, I, F

DCs involved

3,4,6,7,9

Objectives:

This WP builds trust through human-in-the-loop approaches, providing users with tools to validate and refine AI outputs while ensuring usability, co-creation and alignment with real-world workflows and societal requirements.

Task 3.1 – Establishment of Human-in-the-Loop Paradigms:

Co-create human-in-the-loop models enabling users to explore outputs, provide feedback and make corrections. Develop scalable solutions for individual-level and population-level assessments, evaluate workflow integration, and assess usability, reliability and trust. Explainability methods will be integrated in collaboration with WP1.

Task 3.2 – Human-AI Collaboration for Professional End Users:

Co-create improved methods for presenting AI outputs to domain experts, focusing on relevance, clarity and integration into decision-making workflows. The resulting communication strategies will be validated in representative operational environments.

Task 3.3 – Human-AI Collaboration Through Monitoring Dashboards:

Review existing approaches for monitoring trustworthy decision-support systems, define a design framework and co-design a population-level monitoring dashboard. Validation activities will assess accuracy, workflow integration and organisational impact.

Task 3.4 – Human-AI Collaboration for End Users and Stakeholders:

Co-create approaches for communicating AI outputs to non-specialist users in order to improve transparency, trust, autonomy and engagement. AI predictions, risks and uncertainties will be translated into understandable and actionable information.

WP Number: 4

WP title: Clinical Adaptation & Case Studies

Start month – End month: M1–M36

Objectives

To adapt, validate and evaluate trustworthy AI models in clinically relevant settings using retrospective and implementation-oriented studies covering cardiovascular medicine, neurodegenerative disease, critical care and medical-device environments.

Task 4.0 – AI Implementation in Clinical Practice

Identify barriers to healthcare AI adoption and establish a shared understanding of implementation requirements across participating healthcare environments.

Activities include:

- workflow analysis,
- stakeholder engagement,
- implementation-readiness assessment,
- shadow-mode deployment planning,
- impact assessment for future clinical adoption.

Evaluation focuses on diagnostic quality, workflow efficiency, trust and usability.

Task 4.1 – Federated Learning Infrastructure Across Clinical Sites

Expand federated-learning capabilities across participating clinical environments to enable privacy-preserving training and evaluation without centralising sensitive patient data.

Activities include:

- federated infrastructure deployment,
- interoperability testing,
- distributed model evaluation,
- multicentre validation.

Task 4.2 – AI Implementation for Amyotrophic Lateral Sclerosis (ALS)

Retrospective evaluation of multimodal causal AI models using:

- ECG,
- EMG,
- echocardiography,
- biomarkers,
- longitudinal clinical records.

Primary objectives:

- progression prediction,
- survival prediction,
- causal discovery across neurocardiological biomarkers.

The task investigates links between cardiac biomarkers, neuromuscular status and disease progression while developing interpretable prognostic tools for multidisciplinary ALS care.

Task 4.3 – AI Implementation for Transcatheter Aortic Valve Implantation (TAVI)

Retrospective multicentre evaluation of multimodal causal AI models in structural heart disease.

Data sources include:

- cardiac CT,
- ECG,

- demographic variables,
- procedural characteristics,
- mortality outcomes,
- pacemaker implantation outcomes.

Objectives include:

- intervention-risk prediction,
- identification of treatment-outcome confounding,
- multicentre validation,
- federated-learning evaluation.

Task 4.4 – AI Implementation for Heart Failure and Wearable Biosignal Data

Evaluate AI systems using longitudinal healthcare pathways that integrate workplace and remote monitoring data.

Data types include:

- smartwatch biosignals,
- ambulatory recordings,
- hospital measurements,
- longitudinal clinical outcomes.

Primary applications include:

- rehospitalisation prediction,
- mortality risk prediction,
- chronic heart-failure management,
- long-term monitoring support.

Task 4.5 – AI Implementation for Critical-Care and Population-Based Case Studies

Adapt and evaluate multimodal AI models using large-scale critical-care datasets.

Target applications include:

- early detection of acute cardiac dysfunction,

- arrhythmia prediction,
- cardiac-arrest prediction,
- mortality-risk assessment,
- intensive-care decision support.

Validation is conducted using large publicly available critical-care datasets and multicentre healthcare records.

Task 4.6 – AI Translation Into Industrial Healthcare Applications

Transfer methods developed in earlier work packages into industrial medical-device environments.

Two primary demonstrators are maintained:

A. Long-Term EEG Technologies

Transfer explainable and causal AI methodologies developed for cardiovascular biosignals toward:

- seizure classification,
- seizure prediction,
- long-term wearable EEG monitoring,
- neurophysiological decision support.

B. Contactless Neonatal Monitoring

Transfer privacy-preserving AI architectures into:

- neonatal incubator environments,
- contactless vital-sign monitoring,
- camera-based physiological sensing.

This includes validation of distributed AI methods in regulated medical-device development workflows.

WP Number: 5

WP title: Coordination and management

Start month – End month: M1 – M36

Lead participant

Academic Beneficiary A

Participants

All beneficiaries

DCs involved

–

Objectives:

To ensure effective coordination and governance of the Doctoral Network. This WP covers project management, monitoring progress and risks, financial and administrative coordination.

Task 5.1 – Project Management:

Oversee the consortium's scientific, administrative, and financial management, including consortium agreements. Monitor deliverables, milestones, risks, and ethics compliance. Report to the funding authority and ensure alignment with programme framework requirements. Chair the steering and supervisory boards.

Task 5.2 – Quality Assurance and Risk Management:

Implement monitoring mechanisms for the progress of work packages and doctoral candidates. Establish risk management procedures, ensure compliance with regulations, and institutional ethical requirements.

WP Number: 6

WP title: Training and Career Development

Start month – End month: M1 – M36

Lead participant

Academic Beneficiary D

Participants

All beneficiaries

DCs involved

DC1-9

Objectives:

Implementation of an impactful training programme at the interfaces of AI, application domain expertise, and education to enhance PhD career prospects and skills development. Monitoring progress toward training goals TG1–6 (see Sec. 1.3.2).

Task 6.1 – Career development plan:

Set-up of personal career development plan (PCDP, see e.g., Section 1.4) for each researcher plus a comprehensive list of them (all beneficiaries including annual updates).

Task 6.2 – PhD committee:

Establishment and implementation of a PhD committee (see Section 2.2) for all DCs comprising supervisors from academia, the non-academic sector, and associated partners (secondment).

Task 6.3 – Training progress monitoring:

Organisation of training events including quality and needs monitoring via feedback mechanism (Supervisory Board).

WP Number: 7

WP title: Impact and Innovation

Start month – End month: M1 – M36

Lead participant

Academic Beneficiary H

Participants

Academic Beneficiary A

DCs involved

DC1-9

Objectives:

Efficient, timely, relevant impact of the project's outcomes on scientific, technological, innovative and societal levels. Implementation of mechanisms to ensure longevity and sustainability of the created outputs.

Task 7.1 – Communication and Impact Strategy (Academic Beneficiaries H, A):

Development and annual update of a strategy for targeted dissemination, exploitation and communication including progress monitoring. Set-up of appropriate external communication channels, webpages, and a knowledge hub, as well as organizing roundtables and minisymposia at international conferences and workshops.

Task 7.2 – Knowledge Transfer (Academic Beneficiary H):

Transform research outcomes into innovations, guidelines, and policy inputs through structured exploitation measures. Develop and apply a Research-to-Innovation (R2I) Canvas to map the scientific, technological, and societal value of each DC's work and

ensure early alignment with industrial needs and policy priorities.

Task 7.3 – Post-project exploitation (Academic Beneficiary H):

Develop a follow-up research strategy to foster future collaboration by identifying funding opportunities related to mobility, staff exchanges, and networking. This will support scientific collaboration and identify relevant future programme calls for continued joint research activities.

Table 3.1 b Deliverables List (Sanitized)

Scientific Deliverables

Number	Deliverable Title	Short Description	WP	Lead Beneficiary	Type	Dissemination	Due
D1.1	Fairness Benchmark Report	Methodology and metrics for benchmarking state-of-the-art AI models in healthcare diagnosis, addressing the fairness-performance tradeoff.	1	Academic Beneficiary B	R	PU	M18
D1.2	Sensitivity & Uncertainty Quantification Methodologies	Clinician-facing tools to enable safe human-in-the-loop interaction.	1	Academic Beneficiary D	PDE	PU	M18
D1.3	Multimodal XAI Framework	A unified software framework integrating explainability, causality, fairness,	1	Academic Beneficiary A	PDE	PU	M24

Number	Deliverable Title	Short Description	WP	Lead Beneficiary	Type	Dissemination	Due
		sensitivity and uncertainty modules developed across the consortium.					
D2.1	Multimodal Data Integration Pipeline Prototype	Initial integration of heterogeneous data sources into a harmonized multimodal pipeline. Demonstration of data fusion and handling of missing data across modalities.	2	Academic Beneficiary C	PDE	PU	M18
D2.2	Robust and Knowledge-augmented AI Prototype	Delivery of robust, physiology-aware AI models integrating domain knowledge and causal reasoning.	2	Academic Beneficiary A	PDE	PU	M18
D2.3	Privacy-preserving Distributed AI Demonstrator	Operational distributed learning framework for privacy-	2	Academic Beneficiary H	PDE	PU	M30

Number	Deliverable Title	Short Description	WP	Lead Beneficiary	Type	Dissemination	Due
		preserving physiological monitoring data, implementing split/edge learning and privacy quantification metrics.					
D2.4	AI-readiness assessment tool for data	Digital tool to assess AI-readiness of local clinical data availability including checklists and a score.	2	Academic Beneficiary G	PDE	PU	M18
D2.5	AI-readiness assessment tool for processing environments	Digital tool to assess AI-readiness of local clinical processing environments including checklists and a score.	2	Academic Beneficiary G	PDE	PU	M24
D2.6	Interoperability Standards and Compliance Recommendations for AI developers	Definition and validation of interoperability and compliance standards for AI models, including	2	Academic Beneficiary G	R	PU	M36

Number	Deliverable Title	Short Description	WP	Lead Beneficiary	Type	Dissemination	Due
		standard healthcare data formats and metadata schemas aligned with European health-data initiatives.					
D3.1	Software Library for HITL AI for physiological signals/images	Open-source software library providing a platform for HITL development in medical signals and imaging workflows.	3	Academic Beneficiary B	PDE	PU	M18
D3.2	Implementation of AI HITL for physiological signal analysis	Reference implementation built on D3.1 demonstrating HITL concepts for a standard physiological-signal application.	3	Academic Beneficiary B	PDE	PU	M24
D3.3	Review paper on current HITL concepts	Consortium-wide review paper on current HITL AI concepts in healthcare.	3	Academic Beneficiary B	R	PU	M12

Number	Deliverable Title	Short Description	WP	Lead Beneficiary	Type	Dissemination	Due
D3.4	State-of-the-art of continuous monitoring of trustworthiness	Report on systematic literature survey on continuous monitoring methods for trustworthiness in healthcare AI.	3	Academic Beneficiary F	R	PU	M12
D4.1	Clinical AI Challenges & Shadow-Mode Trial Blueprint	Report on identified barriers and draft design for clinical AI shadow-mode trials.	4	Academic Beneficiary E	R	PU	M6
D4.2	Scientific papers on AI models evaluated in retrospective clinical trials	Reporting scientific findings from retrospective clinical trials on multimodal cohorts, intervention cohorts, wearable-sensor cohorts and population-based critical-care cohorts.	4	All Beneficiaries	R	PU	M12, M24, M36
D4.3	Industrial AI Integration Guide	Report for industry stakeholders	4	Academic Beneficiaries H & B	R	PU	M30

Number	Deliverable Title	Short Description	WP	Lead Beneficiary	Type	Dissemination	Due
		supporting knowledge transfer and integration of AI into products and services.					

Management, Training, Recruitment and Dissemination Deliverables

Number	Deliverable Title	Short Description	WP	Lead Beneficiary	Type	Dissemination	Due
D5.1	Establishing IP management	Establishing an IP management plan integrating consortium provisions on ownership, access and exploitation.	5	Academic Beneficiary A	R	CL	M01
D5.2	Management handbook	Defining governance, administrative procedures, communication protocols, partner roles, decision-making structures and reporting lines.	5	Academic Beneficiary A	R	PU	M03
D5.3	Mobility declaration	Outlining planned researcher mobility and compliance with	5	Each Beneficiary	ADM	CL	M06

Number	Deliverable Title	Short Description	WP	Lead Beneficiary	Type	Dissemination	Due
		eligibility requirements.					
D5.4	Scientific progress reports	Early, mid-term and final scientific reports.	5	Academic Beneficiary A	R	PU	M12, M24, M36
D5.5	Consortium meeting	Mid-term consortium meeting.	5	Academic Beneficiary A	ADM	PU	M18
D6.1	Training progress report	Fellows' achievements, skills development and milestones reached during training.	6	Academic Beneficiary F	R	PU	M12, M24, M36
D6.2	Career development plan (PCDP)	Defining scientific goals, transferable-skill goals, impact goals and career-development pathways.	6	Academic Beneficiary F	R	PU	M12, M36
D6.3	Report on PhD theses	Progress, submission and outcomes of doctoral graduation processes.	6	Academic Beneficiary F	R	PU	M36
D7.1	DEC plan and report	Annual plans and reports on Dissemination, Exploitation and	7	Academic Beneficiary H	R	PU	M12, M24, M36

Number	Deliverable Title	Short Description	WP	Lead Beneficiary	Type	Dissemination	Due
		Communication activities.					
D7.2	R2I Canvas	Annual mapping of potential results, IP and TRL of doctoral candidates' outcomes.	7	Academic Beneficiary H	R	PU	M12, M24, M36
D7.3	European Knowledge Hub	Knowledge hub hosting project curriculum, learning materials, datasets and software libraries, ensuring sustainability.	7	Academic Beneficiary H	PDE	PU	M18
D7.4	Follow-up research strategy	Plan for future collaboration and joint funding opportunities.	7	Academic Beneficiary H	R	CL	M30

Table 3.1 c Milestones List (Sanitized)

Number	Title	Related WP	Lead Beneficiary	Due Date	Means of Verification
M0.1	Website and social-media channels online	7	Academic Beneficiary A	M01	Publicly available website, communication channels and launch announcements

Number	Title	Related WP	Lead Beneficiary	Due Date	Means of Verification
M0.2	Establishment of Supervisory Board, Innovation and Sustainability Committee, and DEC Committee	5	Academic Beneficiary A	M03	Meeting records and governance documentation
M0.3	Recruitment completed	5,6	Academic Beneficiary A	M05	Signed employment contracts and recruitment records
M1.1	Development of explainable AI models for cardiovascular data	1	Academic Beneficiary A	M18	Software prototypes and internal technical reports
M1.2	Fairness, sensitivity, uncertainty and causality module integration and benchmarking	1	Academic Beneficiary B	M24	Scientific publications, benchmark reports and software integration results
M2.1	Multimodal data-integration prototype and recommendations	2	Academic Beneficiaries C and G	M20	Prototype implementation, assessment tools and scientific publications
M2.2	Processing-environment prototypes and recommendations	2	Academic Beneficiaries H and G	M30	Prototype implementation, assessment tools and scientific publications
M3.1	Development of human-in-the-loop AI platform	3	Academic Beneficiary B	M18	Open-source software library and demonstration platform

Number	Title	Related WP	Lead Beneficiary	Due Date	Means of Verification
M4.1	Expansion of the federated-learning network to additional clinical sites	4	Academic Beneficiary E	M18	Federated-learning node installed and successfully tested across participating clinical sites
M4.2	Evaluation of AI models in retrospective clinical studies	4	All beneficiaries	M24	Scientific publications and evaluation reports derived from clinical case studies
M6.1	Personal Career Development Plans established for all doctoral candidates	6	Academic Beneficiary F	M07	Completed and signed PCDP documentation
M6.2	60% of secondments implemented	6	Academic Beneficiary F	M24	Secondment reports and review documentation
M6.3	Establishment of PhD committees	6	Academic Beneficiary F	M10	Meeting minutes and committee records
M7.1	European AI Knowledge Hub operational	7	Academic Beneficiary A	M18	Publicly available platform, learning materials and training resources
M7.2	Dissemination of scientific results	7	All beneficiaries	M24	Scientific publications, conference presentations and dissemination records
M7.3	Follow-up research proposal	7	Academic Beneficiary H	M48	Submission of at least one follow-up collaborative research proposal

Table 3.1 d DC Table (Sanitized)

DC No.	Recruiting Participant	PhD Awarding Entity	Planned Start Month	Duration (months)	Total Duration of Secondments (months)	Total Duration in Non-academic Sector (months)
DC1	Academic Beneficiary A	Academic Beneficiary A	M5	36	12	9

DC No.	Recruiting Participant	PhD Awarding Entity	Planned Start Month	Duration (months)	Total Duration of Secondments (months)	Total Duration in Non-academic Sector (months)
DC2	Academic Beneficiary B	Academic Beneficiary B	M5	36	7	3
DC3	Academic Beneficiary C	Academic Beneficiary C	M5	36	12	12
DC4	Non-academic Beneficiary D	Associated Academic Institution D	M5	36	7	3
DC5	Academic Beneficiary E	Academic Beneficiary E	M5	36	8	8
DC6	Academic Beneficiary F	Academic Beneficiary F	M5	36	9	9
DC7	Academic Beneficiary G	Academic Beneficiary G	M5	36	10	0
DC8	Academic Beneficiary H	Academic Beneficiary H	M5	36	9	9
DC9	Academic Beneficiary I	Academic Beneficiary I	M5	36	8	8

Table 3.1 e Project Risks (Sanitized)

Description of risk	Likelihood	Severity	WPs involved	Proposed risk-mitigation measures
Delay in recruitment; insufficient gender balance among applicants; insufficient number of high-quality applications	Medium	Medium	5	Early-on and targeted job advertisements, specific dissemination channels. Project manager monitors number and quality of incoming applications. If not deemed sufficient after 60% of the intended recruitment period, efforts will be reinforced. Inclusive language used in position advertisements and partners follow equality principles in hiring.
Delayed start of doctoral candidates due to visa issues	Medium	Low	All WPs	Early and transparent communication of expected start dates, strong collaboration with researcher-mobility support services and local authorities for visa procedures, and where possible remote work and virtual access to network-wide training activities.
Drop-out of doctoral candidate before end of contract	Low	Medium	1–4	If in an early stage, draw new candidates from the reserve pool of applications. If in a later stage, assess whether work can be allocated to another position.
PhD thesis not completed before end of contract	Medium	Low	1–4	Ongoing support, supervision and monitoring mechanisms during thesis preparation. Extension possible through

Description of risk	Likelihood	Severity	WPs involved	Proposed risk-mitigation measures
				alternative institutional funding where available.
Doctoral candidates not achieving expected training progress or delayed secondments	Low	Medium	1–4	Clear expectation management, strong supervision mechanisms including PCDP, support on multiple levels, sound training and secondment planning, feedback mechanisms after training events, and high-level graduate-school support.
Project lacking impact	Low	High	7	Detailed and regularly updated dissemination, exploitation and communication strategy. Experienced senior scientist as WP leader.
Real-world deployment may reveal limitations in model generalization, infrastructure, or user acceptance	Medium	Medium	4	Adopt a HITL approach with continuous stakeholder feedback to refine models and ensure alignment with real-world workflows, including on-site refinement during implementation activities.
Data sharing between clinical, industrial, and technical partners may be delayed due to additional legal or administrative requirements, potentially slowing model development	Medium	Low	1	Use public, ethically approved datasets and established secure infrastructures to ensure data privacy and compliance while maintaining scientific progress.
Data integration may be challenged by	Medium	Low	2	Apply standardized ontologies and data models, perform

Description of risk	Likelihood	Severity	WPs involved	Proposed risk-mitigation measures
missing values, noise, and heterogeneous data formats. Gains from multimodal fusion may be limited or difficult to interpret				ablation studies to assess the added value of each modality, and validate fusion strategies with domain experts to ensure relevance.
Limited availability of domain experts could slow co-creation, validation, and feedback cycles	Medium	Low	3	Implement flexible engagement formats (e.g., asynchronous feedback) and appoint liaison researchers to coordinate collaboration.

3.2. Quality, capacity and role of each participant, including hosting arrangements and extent to which the consortium as a whole brings together the necessary expertise (Sanitized)

Appropriateness of the infrastructure and capacity of each participating organisation, in light of the tasks allocated to them in the action

The Doctoral Network is supported by a tightly integrated ecosystem of AI, data and clinical infrastructures, fully aligned with the WPs and DC projects. Across the consortium, technical universities, comprehensive universities and academic medical centres contribute complementary, task-aligned facilities.

- **HPC clusters and AI labs (Academic Beneficiaries A, B, C, H):** GPU clusters, dataflow systems and specialised ML/DL/causal-inference toolchains supporting AI model development in WP1.
- **National-scale AI infrastructures (Academic Beneficiary C):** access to large-scale computational resources, AI-driven services and datasets, enabling multimodal health-data research in WP2.
- **Secure medical-data and interoperability infrastructures (Academic Beneficiaries E, G):** secure clinical data facilities and AI pipelines for clinical training and inference in WP2.
- **Clinical registries and biobanks (Academic Beneficiary I):** multimodal registries and biobanks combining clinical, biomarker, biosignal and imaging

data, plus advanced electrophysiology, imaging and neurophysiology facilities supporting WP3–4 use cases.

- **Biomedical measurement and monitoring laboratories (Academic Beneficiary H):** contactless physiological monitoring platforms, simulation environments and multimodal physiological setups supporting privacy-preserving sensing research in WP2.
- **Digital-health, next-generation connectivity and HCI testbeds (Academic Beneficiary F):** campus-wide advanced networking infrastructure, field-testing environments and user-centred digital-health research facilities supporting WP3.
- **Open-access life-science core facilities:** sequencing, flow cytometry, proteomics, structural biology, advanced imaging and biocomputing resources supporting WP4 and providing insight into multi-omics and biomarker pipelines linked to target health applications.

All DCs are embedded in accredited doctoral schools and established research groups with clear hosting arrangements. International offices support housing, integration, language learning and practical mobility aspects.

Consortium composition and exploitation of participating organisations' complementarities

The consortium covers the full innovation chain of trustworthy causal AI in health through deliberate complementarities.

Methodological depth at Academic Beneficiaries A, B, C, F and H spans biomedical signal and image processing, causal and explainable AI, medical imaging and digital-health software engineering.

Clinical and data breadth at Academic Beneficiaries E, G and I includes cardiovascular medicine, electrophysiology, neurology and rare-disease applications.

Academic Beneficiaries G and H contribute **regulatory and interoperability expertise**, including privacy-preserving AI, data-protection standards and compliance with major European regulatory frameworks.

Academic Beneficiary F strengthens the consortium with leading **digital-health and human-technology-interaction capacities**.

These complementarities are embedded in the network's design: each DC project combines at least two environments through co-supervision and international/intersectoral secondments.

Foundational AI outputs flow into infrastructure and deployment activities and then into clinical and digital-health pilots, such as prognostic modelling, intervention-risk modelling and physiological-monitoring applications.

Furthermore, cross-cutting training in open science, ethics, regulation and innovation enables DCs to link technical advances with clinical, legal and implementation needs.

Role of associated partners to the programme

Associated partners strengthen the network with specialised causal-AI expertise, industrial translation capacity and broader clinical validation.

Non-academic Beneficiary D contributes advanced causal and explainable AI expertise and model uncertainty quantification competence, co-supervising DCs and providing access to specialised development environments and datasets.

Industrial Partners A, B and C provide quality-managed development environments, regulatory expertise and facilities for integrating trustworthy AI into real-world products during DC secondments.

Additional associated academic and clinical partners contribute complementary patient cohorts, datasets and care settings for external validation and for human-in-the-loop co-creation.

They are fully integrated into supervision, training and governance, co-supervising DCs, hosting secondments and contributing to network schools, workshops and advisory structures to ensure that scientific advances align with clinical needs and industrial feasibility.

4. Recruitment strategy (including how the project will strive to adhere to the Code of Conduct for the Recruitment of Researchers) (Sanitized)

The Doctoral Network implements a unified, open and merit-based recruitment procedure that adheres fully to the principles of the European Code of Conduct for the Recruitment of Researchers. Coordinated centrally under Work Package 5, recruitment is overseen by a Recruitment Manager and institutional HR departments to ensure that all doctoral candidates are selected through a transparent, inclusive and internationally comparable process.

All DC positions will be advertised simultaneously on EURAXESS, institutional channels and international scientific networks, with additional dissemination through communities supporting women and underrepresented groups in STEM. Advertisements will be written in inclusive language, present the roles broadly enough to attract diverse applicants, and describe working conditions, supervision arrangements, training

opportunities and mobility requirements. Applications will be collected through institutional HR portals to guarantee GDPR-compliant handling. Calls will remain open for at least one month, with deadline extensions and reinforced outreach if the applicant pool lacks gender balance or sufficient qualified candidates.

Evaluation follows a structured and harmonised procedure. Each DC position is assessed by a Selection Committee made up of the main supervisor, a co-supervisor or expert from another partner institution and an HR representative. Committees aim for gender balance and intersectoral representation, ensuring broad expertise and fair evaluation. After eligibility checks, candidates are scored using shared criteria covering academic excellence, research potential, motivation and fit with the network's interdisciplinary goals. Shortlisted candidates are interviewed online, and committees record decisions in a standardised selection report verified by the Recruitment Manager. All applicants receive outcome notifications, including brief feedback for those not selected.

The network embeds gender equality and diversity in recruitment by monitoring gender distribution in applications, longlists and shortlists. If imbalances arise, the Recruitment Manager triggers additional measures such as broader outreach, extended deadlines or targeted communication. Where feasible, anonymised initial screening will be used to minimise unconscious bias. Recruitment outputs are documented transparently in the Management Handbook and subject to oversight by the Supervisory Board. If a position remains unfilled, the consortium reassesses the advertisement and dissemination strategy, republishes the call with improved targeting and, if needed, adjusts timing or secondment planning. These steps follow established MSCA guidance that unsuccessful recruitment rounds require prompt corrective action.

The recruitment timeline is designed for coordination and clarity. All calls will be launched at project start, the application window will run for at least one month, evaluation and interviews will follow immediately, and appointments are expected between months four and six. This harmonised process ensures a fair, coherent and internationally comparable selection procedure that attracts excellent candidates and provides a solid foundation for the network's research and training goals.

5. Network organisation (Fully Sanitized)

The Doctoral Network is organised through a transparent governance structure integrating scientific leadership, training, impact and sustainability. The key bodies are the **Project Coordinator and Management Team**, the **Supervisory Board (SB)**, the **Dissemination, Exploitation and Communication (DEC) Committee**, and the **Innovation and Sustainability Committee**. Together, these bodies ensure coherent

implementation of the research and training programme, systematic monitoring of progress and risks, and alignment with applicable programme rules and project-management requirements. Detailed procedures (e.g. voting rules and escalation mechanisms) are defined in the Consortium Agreement and Management Handbook.

Project Coordinator and Management Team

The **Coordinating Beneficiary** is responsible for overall scientific, administrative and financial coordination of the Doctoral Network under Work Package 5 ("Coordination and Management"). Supported by project-management staff, the Coordinator:

- Monitors deliverables, milestones, risks and compliance activities across all work packages.
- Ensures timely scientific and financial reporting.
- Organises consortium and review meetings.
- Acts as the main interface between beneficiaries, associated partners and the funding authority.

The Project Manager maintains communication platforms and shared repositories, follows recruitment and secondments, collects reporting inputs and prepares meetings of governance bodies. This support enables supervisors and work-package leaders to focus on research, training and supervision activities while ensuring consistent implementation across institutions.

Supervisory Board (SB)

The Supervisory Board is the principal strategic body for research, training and supervision. It includes one representative from each beneficiary and selected associated partners, ensuring academic, clinical and industrial perspectives are represented.

The SB is responsible for:

- Overseeing the scientific direction and coherence of the research programme.
- Monitoring supervision quality and training implementation, including Individual Training Panels (ITPs), PhD Committees and Personal Career Development Plans (PCDPs).
- Reviewing progress reports and training feedback and proposing corrective measures where necessary.

- Ensuring recruitment, mobility and supervision respect researcher-charter principles as well as equality, diversity and inclusion policies.

The SB meets at least twice per year and may convene additional meetings when strategic decisions are required. Doctoral Candidate representatives may participate in discussions relating to training quality, wellbeing and career development.

Dissemination, Exploitation and Communication (DEC) Committee

The DEC Committee is the specialised body responsible for impact-related activities. It includes representatives from academic, clinical, industrial and regulatory sectors together with a doctoral-candidate representative.

Its responsibilities include:

- Designing and annually updating the dissemination, exploitation and communication strategy.
- Coordinating impact activities with Work Package 7.
- Defining and monitoring key performance indicators, including publications, software releases, policy outputs and outreach activities.
- Coordinating external communication channels such as websites, social media, workshops and conference events.

Within the DEC governance structure, an **IP and Exploitation Panel** reviews invention disclosures, advises on licensing and innovation opportunities and ensures that intellectual-property management remains aligned with consortium agreements and programme requirements.

The DEC Committee reports regularly to the Supervisory Board and recommends corrective measures whenever impact indicators require improvement.

Innovation and Sustainability Committee

The Innovation and Sustainability Committee focuses on sustaining impact beyond the project lifetime. It works closely with the Supervisory Board, DEC Committee and relevant work packages to:

- Monitor indicators of long-term collaboration and innovation.
- Support integration of successful training modules into local doctoral curricula.

- Coordinate development of follow-up research and training proposals.
- Ensure sustainability considerations are incorporated into planning and governance.

The Committee supports the long-term sustainability of project outputs, training resources and collaborative structures established by the Doctoral Network.

Monitoring, Decision-Making and Internal Communication

Monitoring and decision-making processes are embedded throughout the governance structure:

- Work Package leaders submit regular progress updates to the Coordinator and Project Manager.
- Training and supervision progress are monitored through annual reports and PCDP reviews.
- Impact indicators and intellectual-property matters are reviewed through the DEC structure and escalated where strategic decisions are required.

Internal communication is supported through:

- Regular virtual meetings.
- Mailing lists.
- Shared collaboration platforms.
- Internal document repositories.

At least one annual network-wide meeting brings together beneficiaries, associated partners and doctoral candidates, typically aligned with major training events to optimise efficiency.

Internal Communication Strategy

The consortium maintains:

- Shared intranet and document repositories.
- Project-wide mailing lists.
- Standard video-conferencing channels.

The Supervisory Board reviews scientific progress, training, risks and finances on a regular basis. Work Package leaders organise online coordination meetings, supported by dedicated communication channels for doctoral candidates. All formal meetings generate concise, action-oriented minutes stored in shared repositories.

Financial Management Strategy

Financial management is embedded within Work Package 5 and governed by the Grant Agreement and Consortium Agreement.

The Coordinating Beneficiary:

- Receives and distributes project funding according to the agreed budget.
- Monitors expenditure across work packages and doctoral-candidate positions.
- Oversees annual internal financial reporting.

Each beneficiary manages the resources associated with its own doctoral candidates and project activities. Where doctoral programmes extend beyond the funded period, institutions may provide complementary resources to support successful completion of doctoral training.

Gender Equality and Diversity in Network Organisation

Gender equality and diversity principles are embedded throughout the governance structure.

The consortium will:

- Strive for balanced representation within governance bodies.
- Consider diversity in committee membership, leadership positions and invited speakers.
- Monitor participation and representation indicators using anonymised and legally compliant statistics.
- Review these indicators regularly and implement corrective actions when persistent imbalances are identified.

Governance bodies may therefore introduce targeted measures to strengthen representation and visibility of underrepresented groups where appropriate.

6. Ethics Self-Assessment (Sanitized)

The project develops trustworthy, causal AI for healthcare applications using retrospective clinical data and limited prospective human-factors studies.

All activities will:

- Follow applicable ethical principles, international standards and national regulations.
- Involve voluntary participation with informed consent where required.
- Exclude activities involving embryos, embryonic stem cells, human tissues, animal experimentation or other sensitive biological research areas.

The principal ethical risks relate to:

- Privacy and data protection.
- Potential re-identification of individuals.
- Bias in datasets, algorithms or outcomes.
- Misinterpretation of AI-generated outputs.

These risks will be mitigated through:

- Pseudonymisation and privacy-preserving data-management approaches.
- Secure processing environments.
- Transparent communication regarding model limitations.
- Appropriate human oversight throughout development, validation and deployment.

AI development follows an ethics-by-design approach aligned with applicable data-protection, medical-device and AI-governance frameworks. Fairness, transparency, robustness and bias-mitigation measures are embedded throughout model design and evaluation. Federated and privacy-preserving methodologies are used where appropriate to minimise unnecessary data movement.

For any international collaboration, all data-related activities will comply with relevant legal requirements governing cross-border data transfers.

All activities involving human participants, personal data or clinical information will commence only after the required ethical, regulatory and data-protection approvals have been obtained, ensuring protection of participants' rights, dignity and safety throughout the project.

The project will comply with the highest ethical standards and with applicable international, European and national legislation concerning:

- Data protection and privacy.
- Research integrity.
- Clinical research.
- Medical technologies.
- Artificial-intelligence governance.
- Health-data governance frameworks applicable during the project lifetime.

Compliance activities will be monitored continuously throughout the project to ensure that scientific excellence is achieved while maintaining ethical, legal and societal responsibility.

7. Environmental aspects in light of the MSCA Green Charter (Sanitized)

The project will implement its research and training in line with the principles of the MSCA Green Charter, encouraging sustainable thinking in research management and day-to-day project practice. The consortium's direct environmental footprint is moderate, as activities primarily involve computational research and collaboration rather than laboratory-intensive work. Environmental measures therefore focus on:

1. Responsible research and computing.
2. Awareness and training.
3. Sustainable organisation of activities.

Environmentally responsible research and computing

Partners will prioritise compact and energy-efficient computational approaches, careful experimental design and extensive code and data sharing to avoid unnecessary retraining and duplication of work.

Where possible:

- Computation-intensive tasks will be carried out using institutional or national computing infrastructures that follow sustainability or renewable-energy strategies.
- Software, datasets and teaching materials will be openly shared to maximise reuse and avoid unnecessary repetition of development efforts.

- Any equipment purchases will, where feasible, follow institutional green-procurement policies and prioritise durable and energy-efficient solutions.

Awareness and training for doctoral candidates and supervisors

Environmental sustainability will be integrated into the transferable-skills programme.

At least one network-wide training event will include topics such as:

- Environmental impacts of AI and digital-health technologies.
- Energy consumption associated with model training, storage and digital services.
- Hardware life-cycle considerations.
- Practical approaches to energy-efficient computing, including profiling, optimisation and resource-aware scheduling.

Doctoral candidates will be encouraged to consider environmental factors in their Personal Career Development Plans, including travel choices, remote collaboration practices and use of shared resources.

Supervisors and principal investigators will also receive guidance on the principles of the MSCA Green Charter to ensure alignment across partner institutions.

Sustainable organisation of project activities and events

Network meetings, training events and secondments will be organised to minimise travel-related emissions and waste while preserving the benefits of in-person interaction.

Measures include:

- Conducting routine meetings of governance bodies and work packages primarily online.
- Clustering in-person events where possible to reduce travel frequency.
- Organising major meetings in locations accessible through lower-emission transport options.
- Preferring venues with recognised sustainability policies or environmental certifications.

These measures aim to balance scientific collaboration with environmentally responsible project implementation.

Beneficiary A

General Description

A leading Central European research university with a strong record in artificial intelligence, signal processing, computer science, and doctoral training. The institution participates extensively in European research programmes and provides advanced educational and research environments supporting translational AI and digital-health innovation.

Role in the Project

- Project coordination.
- Leadership of trustworthy and causal AI research.
- Development of physiology-aware and knowledge-augmented AI methods.
- Coordination of the long-term knowledge-hub sustainability activities.

Research Facilities

- AI laboratories.
- High-performance GPU infrastructure.
- Advanced signal-processing environments.
- Specialised computing systems for machine learning and explainable AI.

Expertise

- Explainable AI.
- Biomedical signal processing.
- Physiology-aware AI.
- Medical imaging.
- AI governance and trustworthy AI.

Beneficiary B

General Description

A major Northern European technical university with internationally recognised research excellence in engineering, digital health, and applied AI. The institution has extensive experience in doctoral training and international research collaborations.

Role in the Project

- Leadership of human-in-the-loop AI research.
- Explainability and fairness evaluation.
- Cardiovascular AI development and validation.
- Training activities and international dissemination.

Research Facilities

- High-performance computing infrastructure.
- Health technology laboratories.
- Innovation and entrepreneurship ecosystem.
- Advanced digital-health research platforms.

Expertise

- Medical AI.
- Explainable AI.
- Clinical decision support.
- Biosignal processing.
- Human-AI collaboration.

Beneficiary C

General Description

A large technical university with internationally recognised expertise in biomedical engineering, multimodal imaging, simulation technologies, and healthcare AI.

Role in the Project

- Multimodal data integration.
- Causal cardiovascular risk modelling.
- Medical imaging and explainability.
- Population-scale healthcare AI validation.

Research Facilities

- Advanced biomedical simulation laboratories.
- Large-scale computing infrastructure.
- National AI infrastructure access.
- Medical-imaging research environments.

Expertise

- Medical imaging.
 - Mobile health.
 - Trustworthy AI.
 - Multimodal learning.
 - Cardiovascular risk assessment.
-

Beneficiary D

General Description

A research and technology organisation linked to a world-leading technological university, with extensive participation in international collaborative research programmes and doctoral training initiatives.

Role in the Project

- Sensitivity and uncertainty modelling.
- Causal AI development.
- Explainability research.
- Translation of AI technologies toward clinical application.

Research Facilities

- Biomedical engineering laboratories.
- Physiological-modelling environments.
- High-performance computing facilities.
- Curated biomedical datasets.

Expertise

- Computational physiology.

- Cardiovascular modelling.
 - Explainable AI.
 - Mechanistic modelling.
 - Multimodal biosignal analysis.
-

Beneficiary E

General Description

A major European university with a dedicated institute focusing on artificial intelligence in cardiovascular medicine and federated learning applications in healthcare.

Role in the Project

- Clinical adaptation and validation.
- Federated-learning infrastructures.
- Multimodal AI for cardiovascular medicine.
- Retrospective clinical studies.

Research Facilities

- Cardiovascular AI institute.
- Clinical imaging infrastructure.
- Medical-data repositories.
- Federated-learning platforms.

Expertise

- Cardiovascular AI.
 - Multimodal learning.
 - Imaging analytics.
 - Clinical implementation.
 - Foundation models for healthcare.
-

Beneficiary F

General Description

A Northern European university with leading activities in digital health, healthcare informatics, human-centred technologies and healthcare transformation.

Role in the Project

- Digital-health implementation.
- Trustworthiness monitoring.
- Human-centred design.
- Career-development leadership.

Research Facilities

- Digital-health test environments.
- Human-experience laboratories.
- Advanced networking and communication infrastructures.
- Health-data innovation platforms.

Expertise

- Digital therapeutics.
- Healthcare informatics.
- Human-centred AI.
- Digital-health governance.
- Healthcare implementation science.

Beneficiary G

General Description

One of Europe's largest university medical centres, combining clinical care, research infrastructures and large-scale health-data environments.

Role in the Project

- Interoperability and AI readiness.
- Clinical informatics.
- Secure data-processing infrastructures.

- Federated healthcare AI.

Research Facilities

- Medical data integration centre.
- Clinical data warehouse.
- Secure computing infrastructure.
- Research data services.
- Healthcare interoperability platforms.

Expertise

- Medical informatics.
 - Health-data interoperability.
 - Clinical AI deployment.
 - Biosignal analysis.
 - Data-governance frameworks.
-

Beneficiary H

General Description

A major European technical university specialising in biomedical engineering, advanced sensing technologies, embedded systems and medical-device development.

Role in the Project

- Privacy-preserving AI.
- Contactless monitoring.
- Medical-device translational research.
- Innovation and impact activities.

Research Facilities

- Biomedical measurement laboratories.
- Contactless monitoring environments.
- Neonatal-monitoring research facilities.
- GPU-based AI infrastructure.

- Secure health-data infrastructures.

Expertise

- Physiological monitoring.
 - Edge AI.
 - Embedded medical systems.
 - Privacy-preserving learning.
 - Medical-device integration.
-

Beneficiary I

General Description

An internationally recognised academic medical centre with strong programmes in cardiovascular medicine, neurology, digital health and AI-supported clinical research.

Role in the Project

- Rare-disease AI applications.
- Clinical adaptation studies.
- Federated validation.
- Human-in-the-loop evaluation.

Research Facilities

- Rare-disease registries.
- Clinical biobanks.
- Electrophysiology laboratories.
- Neurophysiology infrastructure.
- Longitudinal multimodal datasets.

Expertise

- Cardiovascular medicine.
- Neurology.
- Clinical AI implementation.
- Prognostic modelling.

- Digital-health governance.
-

Associated Partner A

Medical Device Manufacturer

Provides expertise in neonatal monitoring systems, medical-device prototyping and regulatory pathways.

Contribution

- Hosts industrial secondments.
 - Supports validation of privacy-preserving physiological-monitoring technologies.
 - Provides prototype-development environments.
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Associated Partner B

Neurotechnology Company

Specialises in ultra-long physiological monitoring systems and wearable neuro-monitoring technologies.

Contribution

- Hosts industrial secondments.
 - Provides expertise in medical-device AI integration.
 - Supports translation of AI methods into clinical monitoring devices.
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Associated Partner C

Emergency and Critical-Care Device Company

Developer of safety-critical monitoring systems and emergency-care technologies.

Contribution

- Supports validation of human-in-the-loop AI.
- Provides device-testing environments.
- Contributes expertise in quality assurance and regulated AI deployment.

Associated Academic Partner D

European Research University

Provides expertise in secure computing infrastructures, trustworthy AI, and doctoral education.

Contribution

- Co-supervision.
- HPC and secure-computing expertise.
- Research on AI reliability and sensitive-data processing.

Associated Academic Partner E

Technological Research University

Provides complementary expertise in biomedical engineering, explainability and uncertainty quantification.

Contribution

- Hosts secondments.
- Supports causal AI and explainability research.
- Provides specialised biomedical-engineering infrastructure.